

(TR-104) PYTHON WITH AI/ML

Training Day 67 Report

Date: 14 April 2026

Topic Covered: AI Video Generation, Stable Diffusion & Automated Instagram Reel Generation

The sixty-seventh day of training focused on AI-powered video generation techniques using Stable Diffusion, Hugging Face models, and automated content generation workflows. The training also included implementation of frame-based video generation pipelines, cloud-based text-to-video generation, and integration of multiple AI systems into a unified Instagram Reel generation workflow.

- Implemented Stable Diffusion pipeline using runwayml/stable-diffusion-v1-5 for high-quality AI image generation.
- Configured Stable Diffusion model to run on CPU-based systems for compatibility without GPU dependency.
- Generated multiple AI-generated frames from a single text prompt for video creation tasks.
- Converted generated PIL images into NumPy arrays for image processing operations.
- Used OpenCV for saving generated frames and assembling them into video files.
- Configured frame resolution and FPS settings for stable video generation output.
- Built a complete frame-to-video pipeline using AI-generated images.
- Integrated Hugging Face InferenceClient with Replicate provider for cloud-based inference workflows.
- Generated videos using Wan-AI/Wan2.2-TI2V-5B text-to-video model through hosted APIs.
- Automated saving and handling of generated .mp4 video outputs.
- Compared local Stable Diffusion workflows with API-based text-to-video generation approaches.
- Integrated Tavily Search API for fetching real-time Instagram trend-related information.

- Processed and structured retrieved web search results for LLM context generation.
- Developed an AI-powered Instagram Reel Script Generator using Ollama (qwen2.5:3b).
- Implemented advanced prompt engineering for generating structured reel scripts and captions.
- Added fallback mechanisms for handling failed or incomplete LLM outputs.
- Extracted structured video prompts from generated scripts for downstream video generation tasks.
- Generated Instagram captions using LLM with predefined formatting constraints.
- Integrated Hugging Face Diffusers (text-to-video-ms-1.7b) for AI-based video generation workflows.
- Applied DPMSolver scheduler and memory optimization techniques for improved inference performance.
- Converted AI-generated frame sequences into final video output using OpenCV video writer functionality.
- Successfully integrated trend fetching, script generation, caption generation, and AI video generation into a single automated workflow.
- Reviewed poster content and suggested grammar and terminology corrections for improved presentation quality.
- **GitHub Link:**
https://github.com/JaspinderKaurWalia26/video_generation
- **Blog Link:**
<https://medium.com/@jaspinderkaurwalia855/generating-ai-powered-videos-using-stable-diffusion-and-opencv-c4285576d3e6>

Overall, the training strengthened understanding of Stable Diffusion pipelines, AI-based video generation, Hugging Face inference workflows, prompt engineering, OpenCV-based video processing, and automated AI-powered content generation systems.